

Fig5 prototype: strict-path synteny/recombination schematic

Geometry comes from strict primary-path selected_segments.tsv rows; coordinates are native 500 kb assembly-window offsets, not CHM13 projections.

Flow legend PAR1 positive-control flow primary PHR donor flow side fragment low-confidence tail same-chromosome context

1. PAR1 positive control

PAN027 paternal hap2 <- PAN011 father

PAR 151 kb; same-chr 349 kb

source: chrX PAR1 context

PAN011#1 chrX:3251-503251

product/child: PAN027 hap2 chrXp (paternal)

chrX:12265-512265

source: chrY PAR1 donor

PAN011#2 chrY:5839-505839

Evidence blocks use local 0-based half-open offsets from selected_segments.tsv; source labels show native 500 kb assembly windows.

2. PHR candidate 1: PAN027 chr9q with chr3q donor

PAN027 paternal hap2 <- PAN011 father

primary donor 45 kb; side fragments 8 kb; low-conf. tail 493 bp; same-chr 446 kb

source: chr9q same-chromosome context

PAN011#1 chr9:135595959-136095959

PAN011#2 chr9:133240004-133740004

product/child: PAN027 hap2 chr9q (paternal)

chr9:135704825-136204825

source: chr3q primary PHR donor

PAN011#1 chr3:202049714-202549714

PAN011#2 chr3:205075647-205575647

Evidence blocks use local 0-based half-open offsets from selected_segments.tsv; source labels show native 500 kb assembly windows.

3. PHR candidate 2: PAN028 strict chr9q path

PAN028 maternal hap1 <- PAN027 mother

primary donor 34 kb; side fragments 16 kb; same-chr 449 kb

source: chr9q same-chromosome context

PAN027#1 chr9:128564684-129064684

PAN027#2 chr9:135704825-136204825

product/child: PAN028 hap1 chr9q (haplotype1)

chr9:134380985-134880985

source: chr3q primary PHR donor

PAN027#1 chr3:200905557-201405557

PAN027#2 chr3:202048279-202548279

Evidence blocks use local 0-based half-open offsets from selected_segments.tsv; source labels show native 500 kb assembly windows.